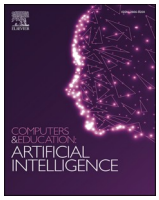
Computers and Education: Artificial Intelligence 6 (2024) 100234



Contents lists available at ScienceDirect

Computers and Education: Artificial Intelligence

journal homepage: www.sciencedirect.com/journal/computers-and-education-artificial-intelligence



Large language models and automated essay scoring of English language learner writing: Insights into validity and reliability

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ARTICLE INFO

Keywords:
Automatic writing evaluation
Automatic essay scoring
Artificial intelligence
Generative AI
Large language model
ChatGPT

ABSTRACT

Advancements in generative AI, such as large language models (LLMs), may serve as a potential solution to the burdensome task of essay grading often faced by language education teachers. Yet, the validity and reliability of leveraging LLMs for automatic essay scoring (AES) in language education is not well understood. To address this, we evaluated the cross-sectional and longitudinal validity and reliability of four prominent LLMs, Google's PaLM 2, Anthropic's Claude 2, and OpenAI's GPT-3.5 and GPT-4, for the AES of English language learners' writing. 119 essays taken from an English language placement test were assessed twice by each LLM, on two separate occasions, as well as by a pair of human raters. GPT-4 performed the best, demonstrating excellent intrarater reliability and good validity. All models, with the exception of GPT-3.5, improved over time in their intrarater reliability. The interrater reliability of GPT-3.5 and GPT-4, however, decreased slightly over time. These findings indicate that some models perform better than others in AES and that all models are subject to fluctuations in their performance. We discuss potential reasons for such variability, and offer suggestions for prospective avenues of research.

1. Introduction

One of the earliest papers on automated essay scoring (AES) laments the plight of English teachers who labor under the burden of "exorbitant" grading responsibilities and how computerized essay scoring could prove a brilliant solution (Page, 1966). 50 years later objections over teacher workload persist, as do dreams of offloading that work to automation (Godwin-Jones, 2022). Furthermore, shifting emphasis on national standards, increased class sizes, and grade inflation have contributed to a decline in student writing ability over the decades (Carter & Harper, 2013; Singleton-Jackson et al., 2009). Teachers are typically reluctant to assign more writing practice as doing so requires the time to read and respond that teachers do not have (Page, 2002). A reliable method of automated scoring of writing seems to be the obvious, yet elusive solution for both reducing teacher workload and allowing for more writing practice for students.

Testing services such as Educational Testing Service (ETS) and exams like the Test of English as a Foreign Language (TOEFL) already employ artificial intelligence (AI) integrated AES tools in their scoring procedures, enabling them to process huge numbers of essays annually. Commercial AES tools, such as *Grammarly,* have also adopted AI to provide formative feedback on writing. However, all of these programs are limited to specific features of writing (e.g., grammatical accuracy, word choice, style) (Ramesh & Sanampudi, 2021), and are typically restricted to academic genres of writing, such as the essay (Warschauer & Ware, 2006). These limitations, in addition to their cost, have stymied their adoption in the classroom.

Recent advances in generative AI may offer a solution to the shortcomings of mainstream AES systems and finally address Page's (1966) complaint of excessive teacher workload, and other drawbacks, to the manual scoring and feedback provision of student writing at the classroom level. Generative AI systems, like OpenAI's GPT-4, are transformer-based deep neural network large language models (LLMs) developed with large amounts of text from the Internet (OpenAI, 2023a). The domain-general nature of the training text for these systems gives them much more versatility in the genres of text they can interact with. Additionally, generative AI is not restricted to specific scoring metrics and can instead analyze a text based on whatever parameters are supplied by the user. Perhaps most importantly, many generative AI tools are free and easy to use, making them accessible to both students and teachers.

Despite the apparent strengths of LLMs, they are not specifically

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https://doi.org/10.1016/j.caeai.2024.100234
Received 14 December 2023; Received in revised form 1 April 2024; Accepted 2 May 2024
Available online 10 May 2024
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designed for AES, nor is it clear how reliable they may be if employed for such purposes. The aim of this paper, therefore, is to explore the validity and reliability of four prominent generative AI LLMs in the scoring of student writing. In better understanding the validity and reliability of generative AI as an AES tool we can open the door to further exploration of this new tool's potential impact on writing assessment.

2. Background & literature

AES involves analyzing written work and applying an evaluation (typically a numerical score) through the use of technology. This has been seen as a solution to human scoring of writing which is not only subject to noise and bias but takes a lot of time to do faithfully (Ramesh & Sanampudi, 2021). The key concerns regarding AES are whether machine scores are valid and reliable, and whether machines ought to be used for scoring human writing.

AES has historically been viewed as a replacement for the human scoring of essays and this has led developers of AES systems to validate AES performance in comparison to human scorers. However, there is not a good understanding of how humans undertake the essay evaluation process (Attali, 2013). For example, Eckes (2008) found that human scorers are inconsistent with which writing features they give weight to when presented with a list of features. Drawing further confusion as to what governs human essay scoring, another study found rater scores to be significantly different even after standardization training (Weigle, 1998). Despite the opaque nature of the human scoring process, AES systems have been able to produce remarkably similar scores with those of human raters (e.g., Page, 1966; Ramineni et al., 2012). However, mainstream AES systems do not read essays, nor analyze them, as a human would (Ericsson, 2006), which invites doubt as to their validity.

The algorithms and models driving mainstream AES tools have become more sophisticated over time, allowing the use of these systems to become more prevalent, particularly in high-stakes exams. Early AES systems, such as Page's (1966) *Project Essay Grade*™, used regression models, and more recent systems such as ETS's *e-rater*® and Pearson's *Intelligent Essay Assessor*™ have employed machine learning techniques (Warschauer & Ware, 2006). These methods all require the preselection of specific features of writing for analysis. For example, grammatical accuracy, lexical variety, and semantic similarity are some common features which are extracted from analyzed essays and compared to a corpus of human-scored essays or defined along other algorithmic parameters (e.g., the ratio of unique word frequency to total words as a measure of lexical variety) (Attali, 2013). In using these machine learning techniques and focusing on certain writing features, machines have been able to reliably produce essay scores that correlate highly with those given by humans (Hussein et al., 2019). However, validity along these terms has been challenged as superficial correlation (Perelman, 2014). If these criticisms are true, validity estimates that compare human and machine scores are tenuous, as humans and machines are likely not measuring the same constructs.

Recent groundbreaking advances in natural language processing and transformer-based AI have opened up new possibilities in AES. According to a recent systematic review, since the public release of generative AI tools, such as ChatGPT by OpenAI in November of 2022, many researchers have speculated that a key affordance of these tools in educational contexts would be for evaluating and scoring student writing (Bahroun et al., 2023). But how they might overcome the validity and reliability shortcomings of predecessor AES systems is yet unknown.

Unlike previous AES systems, generative AI is not specifically trained to score essays and therefore does not rely on analyzing texts for predefined features unless it is specifically prompted to do so. This might make generative AI more similar to humans when scoring writing in that the mechanisms through which it may arrive at an essay score are non-deterministic (Ouyang et al., 2023). Specifically, LLMs use an algorithm of probability, which simply produces the next most statistically likely word in a sequence. The AI sampling techniques create an element of randomness which generates inconsistent output from identical input. This feature has resulted in LLMs demonstrating a high degree of creativity (Guzik, et al., 2023), and rudimentary reasoning ability (Liu et al., 2023). Chan (2023) went so far as to refer to ChatGPT as a nascent form of artificial general intelligence, which describes an AI system that can perform any intellectual task that a human can do. This versatility has already led some researchers to study generative AI's potential for analyzing student writing (e.g., Carlson et al., 2023; Pack & Maloney, 2023a).

The interface of generative AI tools allows users to provide prompts that explicitly outline the implementation parameters for a task. These prompts can be optimized iteratively through prompt engineering (Zhou et al., 2023). Therefore, it is theoretically possible to have ChatGPT, for example, score an essay via the same features that mainstream AES systems use, such as lexical variety or syntactic complexity. Alternatively, it is possible to prompt ChatGPT to provide a score within the parameters that a human rater might use, such as a rubric that reflects a specific curriculum. However, due to the non-deterministic nature of generative AI, the question of whether these systems can reliably adhere to prompt specifications for AES needs to be addressed.

There is also the question of whether machines should be scoring essays at all. Most arguments against AES are contingent on the purpose for the writing. For example, some say that AES deprives writers of an audience and infringes on the social nature of writing (Shermis et al., 2013). Ericsson (2006) argues that writing conveys meaning more artfully than can be conceptualized by an AES algorithm. These arguments have merit, and practitioners should consider carefully the ramifications of submitting student work to AES systems. However, artful writing that is done for the purpose of conveying meaning is one of many motives for writing. Learners of additional languages, alternatively, perform writing as a means of developing language proficiency and are therefore assessed for their proficiency over the quality of their ideas (Weigle, 2013). This type of writing may be more conducive to AES, which is proficient at identifying sentence-level errors in grammar and usage. Exams for language education program placement or workplace certification can benefit from the use of AES to quickly and reliably score essays along proficiency metrics.

Early research into using generative AI for AES and feedback provision have shown positive results. In a preprint by Dai et al. (2023), researchers used ChatGPT to formulate feedback on 102 learner essays and compared the feedback to the instructors. They found that ChatGPT wrote more detailed and easier to read feedback than the teacher did, but struggled to achieve a high level of agreement with the teacher's feedback on certain writing points.

In an experimental study, Mizumoto and Eguchi (2023) used the IELTS task 2 writing band descriptors[1] to score 12,100 TOEFL essays using OpenAI's text-davinci-003 model (a precursor of GPT-3.5) and compared them to human ratings. They concluded that generative AI tools like ChatGPT can accurately assess essays according to the rubric, demonstrating the feasibility and efficacy of generative AI tools for AES.

In a related study, Escalante et al. (2023) employed GPT-4, OpenAI's newest and most powerful model at the time of writing, to provide feedback on English language learner writing. They found that students who received writing feedback generated by AI showed no differences in learning outcomes and performance at the end of the semester from those who received feedback from a human. Additionally, another group of learners' who received both AI and human feedback on their writing were evenly split as to their preference, finding advantages with both. For AI, they remarked on the clarity and specificity of the feedback, for the human feedback, they cited the ability to ask follow-up questions and the higher degree of interactivity. This study demonstrated that

[1] https://takeielts.britishcouncil.org/sites/default/files/ielts_writing_band_descriptors.pdf.

GPT-4 has the ability to analyze ESL student writing and make suggestions on how to improve that writing.

Given the feasibility of using generative AI LLMs like GPT-4 for AES, and their ability to provide actionable feedback, an important next step is in measuring the reliability of LLMs when used for AES. Numerous LLMs have been made available to the public but their comparative capabilities in AES are unknown. Therefore, the purpose of this paper is to explore the reliability of mainstream generative AI LLMs in AES. Understanding how well LLMs perform in prompt-based AES compared to human scorers will provide valuable insight on the efficacy of using these tools for AES as well as broader understanding of the comparative abilities of the prominent LLMs. A key complication in assessing the reliability of LLMs is that models are periodically updated with new training data, and this can affect their performance (Chen et al., 2023). As such, the following research questions guided the study.

1. To what degree do LLMs exhibit intrarater reliability in automated essay scoring?
2. How does the intrarater reliability of the LLMs change over time?
3. To what degree are LLMs' scores valid, as exhibited by interrater reliability with human scores?
4. How does the interrater reliability of the LLMs change over time?

3. Methodology

3.1. Experimental setup and data

The study used data from the admissions test of a small liberal arts university in the Asia-Pacific area. The dataset was composed of essays from a cohort of 119 ESL learners who took a university admissions and placement test to assess language proficiency. Students completed the test online using the Canvas learning management system. Students wrote their answer to an essay prompt within a textbox with access to the learning management system's basic grammar and spellchecker, which functions similar to the grammar and spellchecker in Microsoft Word. Students' essays had an average word count of 287. Standard ethical procedures of the educational institution were adhered to in order to protect the applicants' anonymity.

3.2. Data collection procedures

The mandatory admissions test was created in-house and administered online. As part of the writing portion of this admissions test, applicants were given 30 min to write an opinion essay on whether physical exercise should be a required part of every school day. Each essay was scored, individually by two English for Academic Purposes teachers who used a simple holistic rubric (see Table A.1 in Appendix A). This rubric was created by language educators of the host institution for the purposes of placing incoming students into appropriate levels of English language classes. In total 10 teachers participated in the rating process. All of these teachers hold graduate degrees in Teaching English to Speakers of Other Languages (TESOL), applied linguistics, and/or education, with an average of 18.3 years of experience. The teachers first calibrated on a subset of 10 essays. If any raters had Pearson correlation coefficients below 0.75, they received more training and recalibrated. After calibration, the Pearson's correlation coefficient for the essay scores by two raters was significant at 0.749 ($p < 0.001$), indicating a strong correlation between scores and suggesting reasonable inter-rater reliability.

In addition, four LLMs were used to score the essays. These were Google's PaLM 2 (via chatbot Bard), Anthropic's Claude 2, and OpenAI's GPT-3.5 and GPT-4 (via chatbot ChatGPT). These LLMs are all transformer-based, autoregressive language models that have been trained on large internet corpora. These models were chosen because at the time of data collection they were among the best performing and most widely used models. Although each of these models provides robust language processing abilities, differences in parameters and architecture may lead to variations in quality for certain tasks, such as AES. Because of this, and also due to the saliency of these models, they were selected for analysis.

In line with Ingley and Pack's (2023) prompt engineering guidelines, the selected LLMs were given the same prompt for the essay scoring task (Fig. 1). The prompt assigned a role to the LLM, specified the task, provided criteria for essay assessment (in this case a rubric, available in Appendix A), the writing prompt given to students, and lastly the student's writing. The rubric was typed in a word document, with a brief explanation for each score, for example "Writing with a score of 6 is characterized by the following:" This was followed by a bullet point list of the descriptors for the score band exactly as they are given in Appendix A. The entire prompt was copied and pasted into the chatbots' messaging interface. A new chat was started for each instance we asked the AI tools to assess student writing. Each student's writing was assessed twice by each LLM on two separate instances (henceforth T1 and T2) with a minimum interim of 90 days ($M = 134.3$, $SD = 29.3$; see Table C.1 in Appendix C). No fine tuning, example-driven optimization, or parameter adjustments were performed with any of the LLMs.

3.3. Data analysis procedures

The intraclass correlation coefficient (ICC) was selected to compare inter and intra-rater reliabilities, as recommended for cases using ordinal scales (de Raadt et al., 2021; Shrout & Fleiss, 1979). The ICC is an index of rater error variance commonly used to establish the reliability of measurement tools given a sample of subjects and two or more raters. It has been shown to be an appropriate indicator of agreement among raters even when the assumption of normality is not met by, for example, returning more conservative estimates for skewed distributions (Mehta et al., 2018).

Pearson's correlations were used for comparisons between machine and human scores, as is typical in AES research (Attali, 2013).

For interrater reliabilities, the scores of two human raters for each essay were averaged, as well were the two scores generated by each LLM. Proportional bias analyses (Bland & Altman, 1999) were conducted to further verify the reliability of measurements across instances.

A paired-samples *t*-test was done to compare mean LLM scores at the T1 and T2 scoring instances. Lastly, descriptive statistics for each scoring attempt by instrument are given in Table B.1 in Appendix B. As the LLMs are being developed iteratively, we provide the versions of the LLMs as well as the dates of analysis in Table C.1 in Appendix C. All calculations were made using SPSS 28.

4. Results

The ICC values for interrater reliability and intrarater reliability are given in Table 1. According to Koo and Li (2016), ICC values can indicate excellent (>0.9), good (0.75-0.9), moderate (0.5-0.75), and poor reliability (<0.5). ICC values for interrater comparisons are moderate to good for the T1 scoring instance, with humans and GPT-4 showing the most reliability and PaLM 2 and Claude 2 showing the least. With the exception of GPT-3.5, LLMs improved on interrater reliability for the T2 scoring instance. Intrarater reliability (consistency in scoring for the same rater) was also moderate to good, with GPT-4 again performing the strongest. T2 intrarater values were improved with the exception of GPT-3.5.

Tables 2 and 3 report the proportional bias and limits of agreement for each rater across T1 and T2 rating instances. This analysis can be used to detect whether there is significant disagreement between repeated measures. The non-significant p-values for all paired ratings across T1 and T2 indicates that scores made by the same rater have no proportional bias.

Pearson's correlation coefficients are given in Tables 4 and 5. All correlations were found to be significant at the 0.001 alpha level,

You will be a professional language teacher who is an expert on language assessment and English language placement exams. You will use a rubric to give a score to students' English language learners writing. Each sample of writing will receive a holistic score from 1-6. [characteristics of each score on the rubric inserted here (see Appendix A)]. Here is the writing prompt given to students: "[redacted for test security]". Now apply the rubric given above to give a holistic score between 1-6 to the following student's writing: [student's writing copied here in plain text]

Fig. 1. The prompt used with each LLM.

Table 1
ICC reliability scores for LLMs and human raters.

Rater	Intrarater reliability for T1 (between two scoring attempts using same instrument)	Interater reliability for T1 (between mean scores of LLMs & human, or between two human scores)	Intrarater reliability for T2 (between two scoring attempts using same instrument)	Interater reliability for T2 (between mean scores of LLMs & human)	Intrarater reliability for T1 and T2 (between mean scores of LLMs)
PaLM 2	0.597[a]	0.605[a]	0.702[a]	0.713[a]	0.694[a]
Claude 2	0.836[a]	0.607[a]	0.885[a]	0.701[a]	0.791[a]
GPT-3.5	0.735[a]	0.719[a]	0.605[a]	0.663[a]	0.714[a]
GPT-4	0.897[a]	0.843[a]	0.927[a]	0.779[a]	0.784[a]
Human	N/A	0.855[a]	N/A	N/A	N/A

[a] $p < .001$.

Table 2
Proportional bias analyses for two attempts for T1 using the same instrument.

Rater	Bias	SD	95% Limits of Agreement		*B*	*t*	*p*
PaLM 2	|0.0084|	0.529	−1.03	1.05	0.018	0.157	0.875
Claude 2	|0.084|	0.462	−0.82	0.99	−0.015	−0.202	0.840
GPT-3.5	|0.0084|	0.589	−1.15	1.16	0.076	0.799	0.426
GPT-4	|0.0731|	0.507	−0.92	1.07	0.003	0.043	0.966
Human[a]	|0.1008|	0.643	−1.16	1.36	0.089	1.278	0.204

[a] More than two human raters were involved in scoring.

Table 3
Proportional bias analyses for two attempts for T2 using the same instrument.

Rater	Bias	SD	95% Limits of Agreement		*B*	*t*	*p*
PaLM 2	|0.0588|	0.668	−1.29	1.37	0.055	0.542	0.589
Claude 2	|0.0336|	0.503	−0.95	1.02	0.87	1.390	0.167
GPT-3.5	|0.021|	0.806	−1.56	1.6	−0.002	−0.019	0.985
GPT-4	|0.0126|	0.467	−0.9	0.93	−0.003	−0.053	0.958

Table 4
Pearson's correlation coefficients among rater averages for T1.

	Human	GPT-4	PaLM 2	GPT-3.5
GPT-4	0.731[a]	–		
PaLM 2	0.547[a]	0.646[a]	–	
GPT-3.5	0.604[a]	0.663[a]	0.388[a]	–
Claude 2	0.470[a]	0.613[a]	0.585[a]	0.514[a]

[a] $p < .001$.

however we are primarily concerned with LLM score correlations with human scores. Human ratings correlated highest with GPT-4 in both T1 and T2 scoring instances, although GPT-4's correlation dropped slightly in T2. Claude 2 has the lowest correlation with humans in the T1 instance. However, most correlations were moderate to strong. The correlation between the two human scores is $r = 0.749$ ($p < 0.001$).

Table 5
Pearson's correlation coefficients among rater averages for T2.

	Human	GPT-4	PaLM 2	GPT-3.5
GPT-4	0.638[a]	–		
PaLM 2	0.583[a]	0.595[a]	–	
GPT-3.5	0.515[a]	0.561[a]	0.509[a]	–
Claude 2	0.544[a]	0.605[a]	0.705[a]	0.533[a]

[a] $p < .001$.

Results of paired sample t-tests between T1 and T2 mean scores for each rating instrument are given in Table 6. With the exception of Claude 2, mean LLM scores differed significantly between T1 and T2 scoring instances. Published model updates took place for GPT-4, GPT-3.5, and PaLM 2 but not for Claude 2 during the interim period (see Table B.1 in Appendix B). This may help explain why the scores given by Claude 2 did or did not change significantly over time.

To provide an overall impression of the score dispersions across each rater and scoring instances we have included Fig. 2, which depicts box plots of mean scores. This figure offers two notable points. One is that humans and GPT-4 both tended to use a larger range of rubric scores, with minimum and maximum whiskers extending from 2 to 6. The other is that GPT-4, PaLM 2, and GPT-3.5 were each able to identify some of the same low-scoring outliers, such as student essays 78, 55, and 50.

5. Discussion

This study investigated the validity and intrarater reliability of four salient LLMs (PaLM 2, Claude 2, GPT-3.5, and GPT-4) in scoring ESL student writing. With regard to the first two research questions related to intrarater reliability, GPT-4 was the most reliable at both T1 and T2, demonstrating excellent reliability. Claude 2 demonstrated good reliability, and PaLM2 and GPT-3 exhibited moderate reliability. With the exception of GPT-3.5, LLMs tended to improve over time. As for research questions three and four, which relate to validity, PaLM 2, Claude 2, and GPT-3.5 demonstrated moderate interrater reliability at T1 and T2. The interrater reliability of GPT-3.5 and GPT-4, however, slumped over time. As with intrarater reliability, GPT-4 was the LLM found to be most reliable in scoring similar to the human raters, with good reliability at T1 and T2. GPT-4 had the highest correlation with human raters at T1 (r = 0.731) and T2 (r = 0.638). For comparison, Burstein and Chodorow (1999) reported an average correlation of 0.693 between humans and

Table 6
Paired samples *t*-test between T1 and T2 mean scores for each LLM.

Rater	Instance	M (SD)	t	p	d
GPT-4	T1	4.35 (0.79)	4.07	<0.001	0.69
	T2	4.09 (0.86)			
GPT-3.5	T1	3.81 (0.57)	2.03	0.044	0.57
	T2	3.71 (0.64)			
PaLM 2	T1	4.12 (0.42)	−9.4	<0.001	0.51
	T2	4.56 (0.61)			
Claude 2	T1	4.24 (0.64)	1.7	0.091	0.65
	T2	4.14 (0.74)			

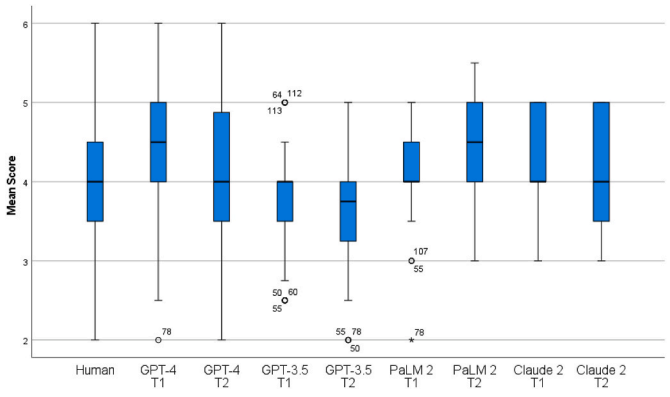


Fig. 2. Box plots for mean scores from humans and LLMs at T1 and T2.

e-rater® on essays written by English language learners, and Bridgeman (2013) mentioned correlations between 0.62 and 0.72 between human and machine scores on TOEFL essays. Therefore, GPT-4's correlation with human raters aligns with the performance of mainstream AES systems when scoring writing from English language learners.

One reason for why GPT-4 demonstrated higher interrater reliability with humans was that it was more willing than the other LLMs to use a fuller range of the rubric scores, as seen in Fig. 2, and by GPT-4's standard deviations which were the highest of the LLMs and comparable to the human scorers' (see Table B.1 in Appendix B). PaLM 2, on the other hand, had the smallest standard deviations and a leptokurtic score distribution at T1 (although, notably, not at T2) indicating a more confined use of the scoring rubric.

While GPT-4 demonstrated high intra and interrater reliability in this study, there are several reasons to be hesitant in applying it, or other LLMs, for AES at this point. To begin with, several of the LLMs exhibited notable quirks when assessing student writing. For instance, GPT-4, GPT-3.5 and PaLM 2, on a few occasions assigned scores on a continuous scale, despite the rubric being ordinal. For another example, there were a few instances where students finished the writing portion of the admissions test with an incomplete sentence lacking a full stop, GPT-4 would respond by finishing the students' paragraph for them and then assign a score, sometimes including its own writing in its evaluation of the students' writing, thereby awarding the writing with a higher score. This was remedied by adding a full stop to the students' writing, which we did before conducting our analyses. Furthermore, as evidenced by their scoring rationale (see Appendix D), the LLMs at times hallucinated features of student writing. For instance, PaLM 2 erroneously claimed that the student "developed 2–3 paragraphs" when the student's writing had no visible paragraphing. It is possible that these quirks can be remedied using few-shot prompting or by adjusting the temperature parameter, and future research should explore these possibilities.

Another reason to be cautious in using LLMs for AES is their non-deterministic nature; the same input given by a user will not generate the same output, a fact that prompted the current study. Several factors lead to this feature of generative AI, such as randomness in the sampling methods and temperature settings (Ouyang et al., 2023). This, plus the frequency at which models are updated, makes it impossible to make an ultimate reliability judgment. Due to this, it is our recommendation that practitioners using LLMs for AES conduct their own reliability analysis specific to their educational context.

The success of using an LLM for assessing writing is also impacted by factors such as the nature of the writing assignment, the rubric utilized for scoring, and the token count required to engineer a successful prompt. As such, the generalizability of GPT-4's success in this study is untenable until further evidence is acquired. For example, we have found GPT-4 to be more successful with simple holistic rubrics, such as the one used in this study that requires only one score as output, than with detailed analytic rubrics with multiple criteria that require a score to be assigned to each criterion. Furthermore, these LLMs have varying limitations on tokens. Generative AI prompts that include instructions to the AI, the rubric, the writing prompt given to students, and the writing of the students may exceed token limits, thereby rendering the AI output less effective and reducing reliability. However, future updates may alleviate the token issue. OpenAI, for instance, has recently released a newer version of GPT-4 (GPT-4 turbo) which affords users with a larger token context window and the ability to create custom versions of ChatGPT called "GPTs" (OpenAI, 2023b). As part of this, users can upload documents that fine-tune the AI for specific tasks. This, and future upgrades, will likely influence the reliability of LLMs for AES and provide ample opportunities for further research on this topic.

Ethical concerns also merit further discussion. Mainstream AES systems have been found to score essays differently along demographic lines (Bridgeman et al., 2012; Ramineni & Williamson, 2018). A valid concern is the potential for LLMs to also perpetuate biases stemming from the material they are trained on. Such AI bias has been found in employment (Bogen, 2019), healthcare (Obenmeyer et al., 2019), criminal justice (Idder & Coulaux, 2021), and education (Baker & Hawn, 2021). Research exploring potential bias of LLMs when tasked with evaluating student writing is needed. It may be possible, for example, that because LLMs are trained on large datasets that may overrepresent certain dialects or varieties of English, they might favor or be more accurate with writing that aligns with these forms. This would potentially put learners who use different structures or vocabulary, reflective of their native language influences, at a disadvantage. Another ethical concern is student privacy (Pack & Maloney, 2023b, 2024). Usage of generative AI for language assessment needs to adhere to institutional and governmental regulations and policies that protect student and participant privacy by, for example, stripping any identifying information provided to LLMs in student writing.

Also worth noting is that some of the arguments against AES persist with the usage of LLMs. For example, some skepticism directed at mainstream AES systems is that they are proprietary, and research into them is restricted. Similarly, the developers of the chatbots and LLMs used in this study, namely OpenAI, Google, and Anthropic, carefully guard the algorithms behind these tools. It is also known that the developers of generative AI do not have a full understanding of the internal workings of LLMs. The high dimensionality of deep neural networks makes it difficult for humans to visualize how generative AI makes decisions or handles new data; a phenomenon known as the black box (Bathaee, 2018). Because of this, a degree of trust is necessary when accepting any AI output.

This study contributes to the research on AES by making the first exploration of the validity and reliability of LLMs for the automated scoring of language learner writing. These new technologies are easy and tempting to use in educational contexts and practitioners should be informed of their limitations. Our findings indicate that certain LLMs might presently be appropriate for limited AES. Their use as a second rater alongside human raters or to resolve a disagreement between two human raters might be acceptable, as might their use in low-stakes or formative writing. If the limitations and ethical considerations presented in this study can be addressed then it is undoubtable that LLMs will see more widespread use in AES.

One limitation of our study is its use of a holistic rubric with a scoring range of 1–6. Small ordinal scoring systems have been found to, at times, exhibit correlations that are more likely to occur by chance (Gardner et al., 2021). With the recent functionality of GPT-4 turbo, future studies could leverage rubrics with more complex scoring systems.

6. Conclusion

The advancement and refinement of LLMs is proceeding at an unprecedented rate. As the technology evolves, we anticipate LLMs will play an increasingly integral role in language assessment, especially if LLMs are fine-tuned for the specific purpose of assessing language. The

next generation of LLMs, such as OpenAI's GPT-5 may prove to be even more capable. In the near future we expect these generative AI tools to be able to reliably, accurately, and rapidly assess hundreds or thousands of writing samples in a short period. Furthermore, we anticipate the trajectory of technological development of LLMs to outpace the rate of academic research and publication, thereby posing both challenges and opportunities for the pedagogical community. While optimistic about this technology, we urge caution in using current LLMs for assessing language. Prior to integrating LLMs into assessment processes, we strongly urge language educators and program administrators to conduct rigorous validity and reliability analyses to ascertain the alignment of the LLM's evaluations with those of human raters and with themselves.

The importance of cross-disciplinary collaboration is increasingly apparent at this intersection of technological advancement and pedagogical practice in the field of language assessment. In particular we see several major research gaps that warrant investigation. First, we need to better understand the capabilities of current LLMs. For example, while GPT-4 was found to be the most valid and reliable model in the current study, further examination of the GPT-4's performance in other contexts that utilize various writing assessments and rubrics would provide insights into the versatility of the LLM and its validity and reliability across contexts. Additionally, the capacity for LLMs to consistently and accurately evaluate multiple students' writing concurrently, by means of uploaded.csv files, ought to be explored. Third, cross-disciplinary collaborative efforts between computational linguists, machine learning experts, and language assessment experts, to fine tune LLMs for the purpose of assessing language would be of particular value as this would likely improve the LLMs' performance in this regard. Furthermore, as the utilization of LLMs for language assessment becomes more widespread, it is of growing importance that the ethical concerns highlighted above are addressed. In particular, it would be advantageous to know if and to what degree LLMs may exhibit bias in their assessment of learners' language. To conclude, it is probable that generative AI tools like LLMs will significantly transform language assessment practices; it is therefore crucial to proactively engage with the potential challenges and opportunities that they present.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Statement on open data and ethics

The data used in this study will be made available upon reasonable request emailed to the corresponding author. The study was approved by Brigham Young University-Hawaii's Institutional Review Board (ID: 23-20). All ethical procedures relating to obtaining participant voluntary consent and data anonymity, as stipulated in the approved IRB application, were carefully adhered to. Participants were protected by hiding their personal during the research and publication processes.

CRediT authorship contribution statement

Austin Pack: Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Alex Barrett:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis. **Juan Escalante:** Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A

Table A.1
Rubric scoring descriptors as inputted into the prompt.

Writing with a score of 6 is characterized by the following:
● Organizes the essay clearly using some cohesive devices accurately.
● Develops 4 or more paragraphs fully.
● Writes developed, concrete supporting detail. Sometimes writes abstract elaborations.
● Uses academic vocabulary appropriately.
● Has infrequent grammar errors that do not affect comprehension; wide range of structures in all time frames, but may not completely control aspect.
● Is easily understood by native readers unaccustomed to non-native writing.
Writing with a score of 5 is characterized by the following:
● Organizes the essay clearly using a few predictable cohesive devices accurately.
● Develops 4 paragraphs.
● Writes concrete supporting detail.
● Uses a moderate amount of academic vocabulary appropriately.
● May have evident error patterns, but they do not distort meaning; uses a variety of time frames and structures.
● Is understood by native readers unaccustomed to non-native writing but requires effort.
Writing with a score of 4 is characterized by the following:
● Organizes some paragraphs clearly, but may not organize the essay clearly or completely. Uses few cohesive devices accurately.
● Develops 2–3 paragraphs.
● Writes personal and sometimes concrete supporting detail.
● Uses sparse academic vocabulary appropriately.
● Has error patterns that may distort meaning or confuse details; uses a variety of time frames.
● Is generally understood by sympathetic readers accustomed to non-native writing.
Writing with a score of 3 is characterized by the following:
● Organizes the paragraph clearly.
● Develops 1 paragraph; other paragraphs may be attempted but are not successful.
● Writes personal supporting detail.
● Uses highly varied general vocabulary.

(*continued on next page*)

Table A.1 (*continued*)

- Has errors that often obscure meaning; uses a limited range of time frames and sentence structures.
- Is generally understood by sympathetic readers accustomed to non-native writing but requires effort.

Writing with a score of 2 is characterized by the following:

- Groups sentences together but does not organize the paragraph clearly.
- Writes at least 5 sentences.
- Writes personal supporting detail within a very limited context.
- Relies heavily on learned phrases; has a limited context for vocabulary.
- Has many errors when creating sentences; uses a very limited range of time frames and sentence structures.
- Is generally understood by sympathetic readers accustomed to non-native writing after rereading.

Writing with a score of 1 is characterized by the following:

- Writes sentences or phrases with minimal grouping.
- Produces a very limited number of sentences or demonstrates an inability to divide sentences.
- Relies almost solely on formulaic/memorized language.
- Has a very limited context for vocabulary.
- Produces insufficient language to assess sentence-level accuracy.
- Is understood with great difficulty even by those accustomed to non-native writing.

Appendix B

Table B.1
Descriptive statistics and Kolmogorov-Smirnov normality statistics for each scoring attempt by instrument.

Rater	Instance	Mean	Standard Deviation	Variance	Skewness	Kurtosis	Kolmogorov-Smirnov Statistic
Human 1	T1	4.160	0.939	0.881	0.299	−0.313	0.257*
Human 2	T1	4.261	0.868	0.754	0.177	−0.279	0.257*
GPT-4	T1A	4.392	0.831	0.690	−0.276	0.166	0.222*
GPT-4	T1B	4.319	0.829	0.687	−0.202	−0.435	0.232*
GPT-4	T2A	4.088	0.894	0.799	0.227	−0.377	0.237*
GPT-4	T2B	4.101	0.896	0.803	0.167	−0.386	0.217*
GPT-3.5	T1A	3.819	0.663	0.440	−0.571	0.704	0.355*
GPT-3.5	T1B	3.811	0.624	0.390	−0.256	0.360	0.342*
GPT-3.5	T2A	3.698	0.757	0.573	−0.146	−0.277	0.269*
GPT-3.5	T2B	3.719	0.758	0.575	−0.366	0.440	0.258*
PaLM 2	T1A	4.126	0.497	0.247	−0.158	2.98	0.415*
PaLM 2	T1B	4.118	0.490	0.240	−0.161	3.2	0.418*
PaLM 2	T2A	4.529	0.711	0.505	−0.321	−0.157	0.301*
PaLM 2	T2B	4.588	0.682	0.465	−0.407	0.027	0.324*
Claude 2	T1A	4.000	0.611	0.373	0.000	−0.256	0.315*
Claude 2	T1B	4.084	0.619	0.383	−0.053	−0.364	0.319*
Claude 2	T2A	4.126	0.819	0.670	−0.332	−1.171	0.252*
Claude 2	T2B	4.160	0.748	0.559	−0.270	−1.161	0.239*

*df 119, $p < 0.001$.

Appendix C

Table C.1
Versions of LLMs utilized and dates analyses were made.

Large Language Model	Chatbot	Instance	Chatbot version	Date of Analysis
Claude 2	Claude	T1	July 11, 2023	August 1–2, 2023
		T2	July 11, 2023	Nov 1–10, 2023
GPT-4	ChatGPT	T1	May 24, 2023	June 8–10, 2023
		T2	Sept 25, 2023	Oct 25–27, 2023
GPT-3.5	ChatGPT	T1	May 24, 2023	June 1–3, 2023
		T2	Nov 6, 2023	Nov 7–17, 2023
PaLM 2	Bard	T1	May 10, 2023	June 1–3, 2023
		T2	Oct 23, 2023	Oct 26–28, 2023

Appendix D. Sample scores and score rationalizations furnished by the LLMs

The mean human score for the associated writing sample was 3.5.

GPT-4

Assessing the provided student writing using the rubric, I would give it a score of 3.
Here's the rationale.

- The essay attempts to organize paragraphs around a central theme, but the organization of ideas is not clear or consistently developed across paragraphs. The use of cohesive devices is minimal.
- There is one clear paragraph developed, with attempts at others that are not fully successful or coherent.
- The supporting details provided are mostly personal and somewhat repetitive, lacking depth or full development.
- The vocabulary used is general and not academic, with some phrases repeated.
- There are frequent grammatical errors, and while some sentences are constructed in different time frames, the errors do obscure meaning and make comprehension effortful.
- The writing can be understood by readers accustomed to non-native English writing, but it requires effort due to the unclear expression of ideas and language errors.

The writing does not fully meet the requirements for a score of 4, which includes more coherent paragraph organization and a better use of academic vocabulary, and it is somewhat above the criteria for a score of 2, where the writer would have more difficulty forming clear sentences and paragraphs.

GPT-3.5

Based on the rubric provided, I would assign the following holistic score to the student's writing:
Holistic Score: 3.
Here's the breakdown of the score based on the criteria.

- Organization: The essay is somewhat organized with some paragraphs, but the organization is not completely clear. There are some cohesive devices used, but they could be improved for better clarity.
- Paragraph Development: The student develops 2–3 paragraphs, but the essay lacks a clear structure, and it's challenging to follow the argument.
- Supporting Details: The writing includes some concrete supporting details but also includes personal opinions and ideas that are not entirely relevant to the topic.
- Academic Vocabulary: There is limited use of academic vocabulary, and some sentences lack precision in their expression.
- Grammar and Errors: There are frequent grammar errors, and some sentences are awkwardly constructed. These errors can sometimes obscure the meaning of the text.
- Comprehensibility: The writing can be understood by readers accustomed to non-native writing with some effort, but it lacks clarity and coherence.

Overall, the writing shows some potential but needs improvement in terms of organization, clarity, and grammar to receive a higher score.

PaLM 2: *Holistic score: 4*

Analysis

The student's writing is organized into some paragraphs, but the essay is not completely organized. The student uses a few cohesive devices, but not accurately. The student develops 2–3 paragraphs fully, but the supporting detail is sometimes personal and not always concrete. The student uses sparse academic vocabulary appropriately, but there are error patterns that may distort meaning or confuse details. The student's writing is generally understood by sympathetic readers accustomed to non-native writing.

Specific feedback

The student should use more cohesive devices to connect their ideas. For example, they could use transition words like "however," "therefore," and "in addition."

The student should develop their supporting details more fully. For example, instead of saying "so we can focus on our education path that we trying to reach," they could say something like "so that we can concentrate in class and learn more effectively."

The student should use more academic vocabulary. For example, instead of saying "something apply to sports and any kind off activity that will put out from other people view opinion," they could say "the same applies to sports and any other activity that distinguishes people from each other."

The student should proofread their work carefully to catch grammar errors. For example, the sentence "so we can achive" should be "so that we can achieve."

Overall, the student's writing is good for a level 4 ELL writer. They are able to communicate their ideas clearly and support their opinion with evidence. However, there are some areas where they could improve, such as using more cohesive devices, developing their supporting details more fully, and using more academic vocabulary.

Claude 2

Based on the rubric, I would give this response a holistic score of 3.

The writing is organized into paragraphs, but the paragraphs are not fully developed. There is an attempt to state an opinion and provide some supporting details, but the ideas are not fully explained.

The vocabulary is quite varied and general. There are many grammar errors that occasionally obscure meaning, such as incorrect verb tenses and forms, missing words, and unclear pronouns. The range of sentence structures is somewhat limited.

The response seems to rely more on personal ideas and experiences rather than concrete details. Academic vocabulary is used sparingly.

While sympathetic readers accustomed to non-native writing may understand the general message, it would require some effort. The errors distract from the flow of the writing.

Overall, this response meets many of the criteria for a score of 3 based on the rubric. The lack of academic vocabulary, limited sentence structures, and frequent errors that obscure meaning prevent it from reaching a 4. With effort, the opinion can be understood, so a 2 would be too low. Therefore, I

would give this response a holistic score of 3.

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